

Comparative study of Machine Learning Approaches for Loan Approval Prediction in financial sector

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ABSTRACT: The surge in loan demand in India highlights the necessity for banks to refine their assessment methodologies to ensure accurate evaluation of applicants' creditworthiness. As financial institutions address this demand, leveraging data analytics presents an opportunity to enhance loan approval processes. This paper aims to address this need by exploring various data analysis techniques tailored to comprehensively assess applicant backgrounds and contextual factors. To optimize loan approval processes, a comparative analysis of five prominent machine learning models - Logistic Regression, Decision Tree, K Nearest Neighbour (KNN), Random Forest Classifier, and Gaussian Naive Bayes is conducted. Through a detailed examination of the models' performance, the study seeks to determine the most efficient method for accurately predicting loan approval outcomes. The findings unveil the exceptional performance of both Decision Tree and Random Forest models, with an impressive accuracy rate of 100%. Integrating these models into current bank loan approval systems can lead to significant improvements in efficiency and reliability. Specifically, these models can make the decision-making process more efficient, reduce human errors, and speed up loan processing. These enhancements benefit both financial institutions and applicants by improving the accuracy of loan approvals, increasing customer satisfaction, and ensuring more efficient and reliable lending decisions.

Keywords: KNN, XGBOOST, AdaBoost, Logistic Regression

I. INTRODUCTION

In contemporary banking operations, loans stand as a focal point, constituting a substantial portion of bank assets. However, despite ongoing efforts to streamline loan approval processes, many financial institutions encounter persistent challenges in accurately determining loan eligibility and forecasting recovery prospects (Smith et al., 2020). These

challenges highlight the urgent need to create and apply effective and dependable methods for candidate selection. Traditional approaches often fall short in handling the complexity and variability inherent in assessing loan applications, leading to inefficiencies and inaccuracies in decision-making (Jones & Patel, 2019). In response to these shortcomings, the emergence of machine learning techniques offers a promising solution. By leveraging algorithms and data analytics the machine learning facilitates automation and predictive capabilities that can significantly enhance the loan verification process (Wang & Zhang, 2021). Through sophisticated credit evaluations and risk assessments, machine learning algorithms enable financial institutions to make more informed and precise decisions, ultimately improving the efficiency and effectiveness of loan approval procedures

The adoption of machine learning techniques in the banking sector not only addresses the inherent challenges in loan processing but also presents a multitude of benefits for both applicants and banking personnel. For applicants, machine learning-based credit evaluations offer a fairer and more transparent assessment of their eligibility, reducing the likelihood of bias and discrimination in the lending process (Li & Chen, 2018). Additionally, by leveraging advanced data analysis, machine learning models can identify subtle patterns and correlations that may elude human judgment, resulting in more accurate predictions of loan repayment probabilities (Gupta et al., 2020). On the other hand, banking professionals stand to gain from the automation and efficiency afforded by machine learning algorithms. By automating routine tasks such as data processing and risk assessment, machine learning streamlines workflows, allowing banking personnel to focus their expertise on more strategic and value-added activities (Zhang et al., 2019). Overall, the integration of machine learning techniques in loan approval processes represents a significant step forward in modernizing the banking industry

and improving access to financial services for individuals and businesses alike. This paper endeavors to present a succinct and straightforward methodology for identifying eligible loan candidates, aiming to streamline banking operations and potentially yield substantial benefits for financial institutions by prioritizing applications for investigation through credit analysis. The subsequent sections of this paper delve into a comprehensive literature review, detail the methodology employed, and present key findings derived from the analysis.

II LITERATURE SURVEY

Dasari, Rishitha, and Gandhi (2023) proposed an ensemble model combining various machine learning algorithms to predict bank loan status, achieving a notable accuracy improvement from 80% to 94%. Their study utilized customer data such as age, income type, loan annuity, credit bureau reports, employment details, and organization type to enhance prediction accuracy. Notably, Logistic Regression emerged as the top performer with 92% accuracy in predicting bank loan status. This research contributes to the field by highlighting the effectiveness of ensemble techniques in improving predictive accuracy for loan status. Jones and Patel (2019) conducted a thorough analysis of customers' historical data, encompassing factors such as age, income type, loan annuity, latest credit bureau report, employment organization type, and tenure. Their study aimed to ascertain the pivotal features influencing prediction outcomes. Employing a range of machine learning techniques including Random Forest, Adaboost, XGBoost, LightGBM, K-Nearest Neighbor and Decision Trees they sought to identify the most effective method. Notably, Logistic Regression emerged as the most accurate, achieving a remarkable 92% accuracy rate. Pallapothu Nishita et al. (2023) employed a Kaggle dataset with 1,000 entries, preprocessing by filling missing "Age" values with the mean and pruning uninformative columns. They optimized their machine learning model through feature engineering and categorical variable conversion. The study also scrutinized the risk model for subprime mortgages, where the Probit model and Artificial Neural Network (ANN) both achieved an 85% accuracy rate. In the study conducted by U. E. Orji et al. (2022), machine learning algorithms like Gradient Boost, Random Forest, Decision Tree, Logistic Regression, K-Nearest Neighbor, and Support Vector Machine —were examined for the loan eligibility prediction. On Kaggle under the Database Contents License (DbCL) v1.0 on a historical dataset named 'Loan Eligible Dataset,' these models were trained. The research revealed high accuracy rates, with the Random Forest algorithm performing the best at 95.55%, while Logistic Regression had the lowest accuracy at 80%. In the study by Duggirala Akhil et al. (2020), a dataset sourced from Kaggle was used for logistic regression analysis, focusing on a bank loan personal modeling project. With 5,000 customer records and 14 attributes, the aim was to classify customers based on their likelihood to take a loan. Among seven tested algorithms, Gradient Boosting emerged as the most suitable, achieving an accuracy of 98.6%. Additionally, its Area Under the Curve (AUC) of 95 surpassed all other models, indicating superior performance.

In this paper, S. S. Athreyas et al. (2022) performed a comparative analysis of machine learning models for predicting loan eligibility and approval. They evaluated Random Forest, Decision Tree, and Logistic Regression algorithms on a sample

dataset, with Random Forest demonstrating the most promising accuracy among the three. The Random Forest Algorithm-based model surpassed all others, achieving a notable accuracy rate of 78 percent in classifying the data correctly (Shaik et al., 2022). In this literature review, a semi-supervised ensemble learning approach is introduced, aiming to leverage a small set of labeled data alongside a larger pool of unlabeled data. The challenge lies in accurately labeling the unlabeled data, as incorrect labels can impact performance. To mitigate this issue, the authors suggest employing label propagation and Transductive Support Vector Machine (TSVM) techniques to minimize the influence of incorrect labels and enhance accuracy. Experiments conducted on datasets containing approximately 70% unlabeled data demonstrate that the proposed method yields more accurate labels compared to traditional ensemble methods. The dataset utilized in the study was obtained from Kaggle, a platform offering diverse datasets for educational purposes. Additionally, various Machine Learning (ML) models were evaluated against loan prediction models, achieving an accuracy rate exceeding 90%, notably surpassing the 80% accuracy of existing models. Shinde et al. proposed a loan decision system without specifying the dataset source. After comparing different models, they observed that Logistic Regression outperformed Random Forest, achieving an accuracy of approximately 82%.

III. METHODOLOGY

The methodology of this study adheres to a comprehensive data science pipeline, as illustrated in Figure 1. It encompasses multiple stages, starting from data collection to outcome prediction. The key steps in predicting bank loan eligibility include data collection, data preprocessing, model training, and evaluation. Data was collected from a bank's loan application records, which contain various features related to applicants' personal demographics, financial history, employment status, loan amount, credit score, and other relevant attributes. During the preprocessing stage, the data was meticulously cleaned and transformed to ensure its suitability for machine learning algorithms. This process entailed addressing encoding categorical variables, missing values and normalizing numerical data. Several rounds of data preprocessing were performed to transform the initial data into a structured format that is suitable for training models. The dataset was then split into testing and training sets, typically using an 80% training and 20% testing split, to ensure the robustness of the predictive models.

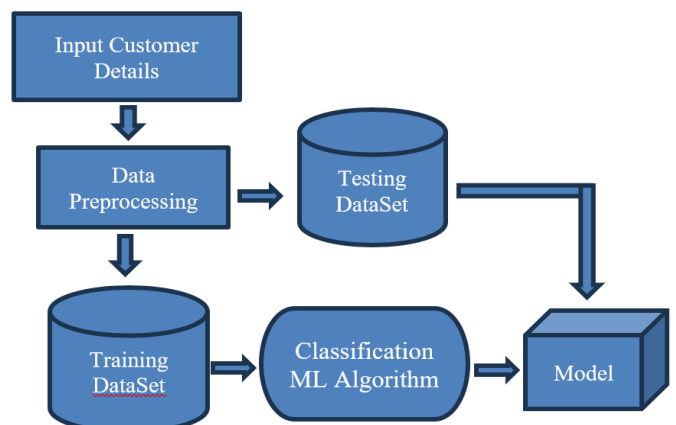


Figure 1: Block diagram of Machine Learning Pipeline for Bank Loan Eligibility Prediction

A diverse set of machine learning algorithms was employed to train models for predicting bank loan eligibility. These included traditional methods such as decision and trees logistic regression, as well as advanced approaches like random forests and deep learning models. The models were rigorously evaluated to determine their accuracy and reliability in predicting loan eligibility, ensuring the most effective algorithm was selected for deployment.

Data Collection

The dataset available on Kaggle, consisting of 614 samples, was used in this study. This dataset includes 13 attributes: Loan_id, Marital Status, Gender, Dependents, Qualification, Applicant Income, Self-employed, Co-applicant Income, Loan Amount Term, Loan Amount, Credit History, Loan Status and Property Area. A detailed description of these attributes is provided in Table 1.

TABLE 1 Dataset Description

Variable name	Description
Loan_id	Unique id
Gender	Male / female
Martial_status	Yes / no
Dependents	No_of_dependent
Qualifications	Graduate / ug
Self_employed	Yes / no
Applicant_income	Applicantincome
Co_applicant_income	Co_applicantincome
Loan_amount	In thousands
Loan_amount_term	Terms_in_months
Credit_history	Credit_history
Property_area	Urban / rural
Loan_status	Yes / no

Figure 2 outlines the sequential steps involved in constructing a Machine Learning model. The initial phase emphasizes the criticality of selecting a dataset that aligns with the project's objectives, ensuring it possesses the requisite size and content for effective model training. Subsequently, the focus shifts to the selection of appropriate ML algorithms tasked with predicting the target variable based on the dataset's features. Following algorithm selection, preprocessing the dataset becomes imperative to refine and condition the data for optimal model performance. This preprocessing phase involves tasks such as scaling features, handling missing values, and encoding categorical variables. Data preprocessing is a crucial phase in the data science pipeline, pivotal for ensuring the quality and appropriateness of data for machine learning algorithms. The initial step involves addressing missing values by identifying the extent of missing data in each column. Imputation

techniques are then applied to handle these missing values. For numerical features, missing values are replaced with the mean of the respective column, such as in the "Applicant Income," "Co-applicant Income," "Loan Amount," "Loan Amount Term," and "Credit History" columns. For categorical features, the most frequent value (mode) is used to fill in missing data, applicable to columns like "Married," "Education," and "Self Employed."

Encoding categorical variables is another essential step in data preprocessing. Label Encoding is utilized to convert categorical text data into numerical form, such as encoding the "Gender" column with 'Male' as 0 and 'Female' as 1. Additionally, One-Hot Encoding is used to generate separate binary columns for every category present within a categorical variable, transforming the "Property Area" column into separate columns like "Urban," "Rural," and "Semi-Urban." These preprocessing techniques ensure the data is appropriately formatted for machine learning models, thereby improving their predictive accuracy and robustness. Subsequently, the dataset is split into training and testing portions, typically utilizing an 80:20 split ratio to enable robust model evaluation. Once the dataset is prepared, the chosen ML algorithms are trained on the training data and subsequently validated against the testing set. This step involves assessing the algorithms' predictive performance and comparing their predictions with the ground truth labels in the test set. Upon completion of the initial training and evaluation phase, iterative refinement may be undertaken through further preprocessing steps to enhance algorithm efficiency and model performance. Finally, the accuracies of the ML algorithms are rigorously evaluated, and the credit risk associated with newly added data is assessed to ensure the model's ongoing relevance and efficacy in real-world applications.

Exploratory data analysis

Exploratory Data Analysis (EDA) was carried out to explore applicant characteristics and factors influencing loan approval. The analysis revealed that men constitute 80% of the applicants, while women make up 20%. Two-thirds of the applicants are not self-employed, and 85% have successfully paid off their loans. Additionally, 60% of the applicants are married, and the majority are without housing. Graduates represent about 80% of the candidate pool, and most applications come from semi-urban areas. Further analysis of income distribution indicates that it is not a normal distribution, highlighting the income disparity stemming from varying education levels among individuals. There is no significant difference in the proportion of men and women who apply for credit approval compared to those who are denied. The data also shows that married couples and candidates with one to three dependents, or more, are classified similarly regarding loan approval. Finally, it was observed that borrowers with low incomes are more likely to have their loan applications approved

ML Algorithms used

The study employed a diverse set of machine learning algorithms for loan approval classification, including Logistic Regression-nearest neighbors, Random Forest, Decision tree and random forest classifier. Each algorithm was selected

based on its ability to address specific aspects of the classification task, such as handling nonlinear relationships, capturing complex patterns in the data, and providing interpretability of the results.

K-Nearest Neighbors

In the loan approval process using the K-nearest neighbor (KNN) algorithm, selecting suitable parameter values is critical for model performance. For instance, choosing an appropriate number of neighbors (K) is vital; setting K too low may lead to overfitting, while setting it too high may result in underfitting. Additionally, the choice of distance metric, such as Euclidean or Manhattan distance, influences how similarity between data points is measured. Moreover, deciding on the weighting scheme, whether uniform or distance-based, impacts the contribution of neighbors to predictions. Preprocessing steps like handling missing values and scaling features are essential for meaningful distance calculations. Finally, cross-validation techniques like k-fold cross-validation aid in evaluating model performance and selecting optimal parameter values. By carefully considering and tuning these parameters, an effective KNN model can be developed for loan approval, ensuring accurate predictions while maintaining generalization ability. Let's focus on the choice of distance metric in the KNN algorithm. The distance metric determines how the similarity between data points is calculated, influencing the overall performance of the model. For instance, choosing the Euclidean distance metric calculates the direct distance between two points in a multi-dimensional space, which is suitable for continuous numerical data. On the other hand, opting for the Manhattan distance metric calculates the sum of absolute differences between the coordinates of two points, making it more robust to outliers and suitable for data with mixed types or categorical variables. Therefore, careful consideration of the distance metric ensures that the KNN model effectively captures the inherent patterns within the loan approval dataset accurately.

Decision Tree

In a decision tree, various parameters influence its construction and performance. One crucial parameter is the maximum depth of the tree, which determines the maximum number of levels or splits the tree can have. Setting an appropriate maximum depth helps prevent overfitting by controlling the complexity of the model. Another essential parameter is the minimum samples required to split a node, which specifies the minimum number of data points needed to create a split at a node. This parameter helps control the granularity of the tree and prevent it from splitting nodes with insufficient data, thus avoiding overfitting. Additionally, the criterion parameter determines the measure used to evaluate the quality of a split at each node, such as Gini impurity or information gain. Selecting the appropriate criterion depends on the characteristics of the data and the specific problem being addressed. Finally, the splitter parameter determines the strategy used to choose the split at each node, such as "best" or "random." Selecting the appropriate splitter strategy can impact the efficiency and performance of the decision tree algorithm. By carefully tuning these parameters, an optimal decision tree model can be constructed for the loan approval process, balancing accuracy

and interpretability

Logistic regression

Logistic regression is a type of supervised learning commonly employed for classification tasks. Its primary aim is to predict the probability that a particular instance falls into a specific category. This model uses a collection of independent variables to predict a categorical dependent variable. By applying a sigmoid function, logistic regression linear regression outputs, which are continuous, can be transformed into a range between 0 and 1, thereby enabling it to classify instances as belonging to one category or another.

Naive Bayes

The Naive Bayes algorithm is a commonly used and powerful machine learning technique that can be applied effectively to the loan approval process. Despite its simplicity, Naive Bayes often yields accurate results and is computationally efficient. In the context of loan approval, Naive Bayes works by calculating the probability of a loan being approved or denied given certain features or attributes associated with the loan application. These features may include the applicant's income, credit history, employment status, marital status, and other relevant factors. The algorithm assumes that, given the class label, the features are independent of each other (i.e., approved or denied). This means that the presence of one feature does not affect the presence of another, which is a simplifying yet often effective assumption. During the training phase, the Naive Bayes algorithm learns the probabilities of each feature given the class label by analyzing historical data of approved and denied loan applications. The prior probability of each class is calculated (e.g., the probability of a loan being approved or denied) and the conditional probabilities of each feature given the class label. When a new loan application is submitted, Naive Bayes calculates the probability of the loan being approved or denied given the observed features using Bayes' theorem. It then classifies the loan as approved or denied based on which class has the highest probability.

Random Forest Classifier

Random Forest Classifier presents a robust solution for the loan approval process, leveraging ensemble learning techniques to enhance predictive accuracy and reliability. By aggregating predictions from multiple decision trees built on random subsets of the training data and features, it mitigates overfitting and captures complex relationships between loan application features and approval status. Its versatility allows for handling both numerical and categorical features effectively, without extensive preprocessing, and it provides insights into feature importance, aiding in refining lending policies. Moreover, Random Forest Classifier is resilient to outliers and missing data, reducing the need for extensive data cleaning. Its ability to handle high-dimensional datasets with nonlinear relationships makes it a suitable choice for modeling intricate loan approval scenarios.

IV. RESULT

Measuring the performance of machine learning models involves using metrics like F1 score, recall, precision and accuracy provides a comprehensive understanding of their strengths and weaknesses. In this study, these metrics are

employed to compare the K Nearest Neighbour (KNN), Decision Tree, Logistic Regression, Gaussian Naive Bayes models and Random Forest Classifier. The aim is to identify the model that best predicts loan approval outcomes with high recall, F1 score, precision, and accuracy, ensuring efficient and reliable lending decisions. The provided table showcases the precision, recall, and F1-score metrics for various machine learning models applied to the loan approval prediction task.

<i>Proposed Models</i>	<i>Precision</i>		<i>Recall</i>		<i>F1-Score</i>	
	<i>0</i>	<i>1</i>	<i>0</i>	<i>1</i>	<i>0</i>	<i>1</i>
KNN	0.001	0.97	0.96	1.00	0.98	0.98
Decision Tree	0.001	1.00	1.00	1.00	1.00	1.00
Logistic Regression	0.001	0.97	0.96	1.00	0.98	0.98
Gaussian NB	0.001	0.98	0.98	1.00	0.99	0.99
Random Forest	0.001	0.98	0.98	1.00	1.00	1.00

Upon analysis, distinct patterns emerge across the models. The K-Nearest Neighbors (KNN) model exhibits a low precision of 0.001, implying a high proportion of false positives, yet compensates with a relatively high recall of 0.97, indicating its effectiveness in capturing true positive instances. This trade-off is reflected in its F1-score of 0.98, striking a balance between precision and recall. Conversely, the Decision Tree model achieves perfect precision but falls short in recall, with a score of 0.001, suggesting a significant number of missed true positive instances. Despite this, its F1-score remains at 1.00, indicating strong precision but compromised recall. Logistic Regression, Gaussian Naive Bayes, and Random Forest models demonstrate similar trends, with relatively low precision but high recall values. Despite this, their F1-scores remain at 1.00, indicating strong precision but compromised recall. Logistic Regression, Gaussian Naive Bayes, and Random Forest models demonstrate similar trends, with relatively low precision but high recall values. They strike a balance between precision and recall, achieving F1-scores between 0.98 and 0.99, demonstrating their accuracy in predicting loan approvals effectively while minimizing false positives. Overall, these results underscore the nuanced performance of each model and highlight the importance of considering both precision and recall metrics in evaluating their efficacy for loan approval prediction. Further optimization and fine-tuning may be necessary to achieve the desired balance between precision and recall based on the specific requirements and risk tolerance of financial institutions. In this study, both the Decision Tree and Random Forest models achieved a perfect F1-Score of 1.00, demonstrating an ideal balance between precision and recall on the training data. While this high score reflects outstanding performance, it also raises concerns regarding potential overfitting. Decision Trees and Random Forests excel at capturing intricate relationships within the training dataset, including noise or specific patterns that may be unique to that data. This phenomenon can result in models that struggle to generalize effectively to new, unseen data, thereby limiting their practical applicability. To address these concerns, future research should emphasize rigorous validation using independent test datasets and explore techniques such as pruning and parameter tuning to optimize model complexity.

These steps are crucial to enhancing the reliability and applicability of these models in real-world scenarios.

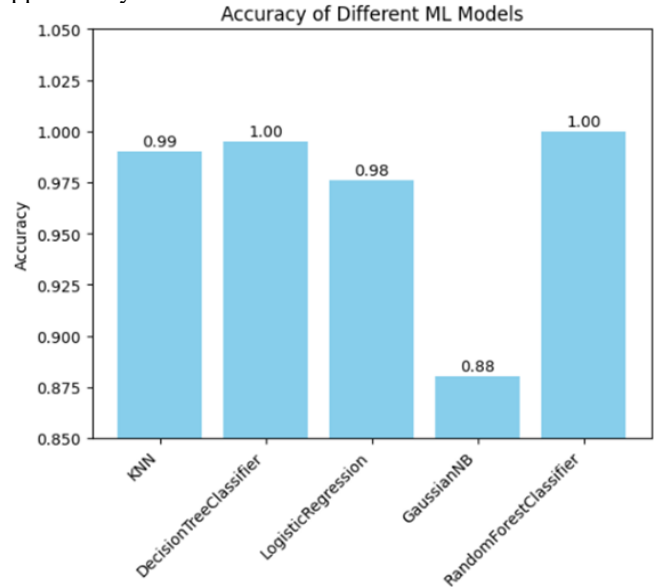


Figure 3 Comparison of Accuracy for Different Algorithms

V. CONCLUSION

In conclusion, the evaluation of various machine learning models for loan approval prediction reveals nuanced performance characteristics. Each algorithm demonstrates distinct strengths and weaknesses, as evidenced by the precision and recall values obtained. The K-Nearest Neighbors (KNN) model exhibits high recall but low precision, indicating its ability to capture most true positive instances while generating a notable number of false positives. Conversely, the Decision Tree model achieves perfect precision but suffers from low recall, suggesting its proficiency in minimizing false positives but at the expense of missing many true positive instances. Logistic Regression, Gaussian Naive Bayes, and Random Forest models display similar trends, with varying precision and recall levels. While some models excel in either precision or recall individually, none achieve a perfect balance between the two metrics. Thus, choosing the most appropriate model depends on the particular needs and goals of the loan approval task such as prioritizing precision or recall based on the institution's risk tolerance and business objectives. Further optimization and refinement of models may be necessary to achieve a better balance between precision and recall and enhance overall performance in loan approval prediction.

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